

Problem Solution Fit

Date	07 July 2026
Team ID	xxxxxxx
Project Name	Voice-Based Concept Understanding Analyser (VBCUA).
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

The **Problem-Solution Fit Canvas** is a strategic, one-page template used to validate if a proposed product or service genuinely solves a target audience's needs. It bridges the gap between customer discovery and product development, preventing from building things nobody wants.

1. CUSTOMER SEGMENT(S) [CS]

- College Students
- Teachers / Faculty Members
- Educational Institutions
- Online Learning Platforms

6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL]

- Limited access to advanced AI assessment tools.
- Internet connectivity issues in some locations.
- Different accents and speaking styles.
- Limited experience with AI-based educational platforms.

5. AVAILABLE SOLUTIONS PROS & CONS [AS] Traditional Viva

- **Pros:** Direct interaction with students.
- **Cons:** Time-consuming, subjective, inconsistent scoring.

Written Exams

- **Pros:** Easy to conduct for large groups.
- **Cons:** Cannot effectively measure verbal conceptual understanding.

Online Quiz Platforms

- **Pros:** Fast and automated.
- **Cons:** Focus mainly on objective questions rather than spoken explanations.

2. PROBLEMS / PAINS + ITS FREQUENCY [PR]

- Manual evaluation of viva and oral assessments is time-consuming.
- Traditional written exams mainly test memorization instead of conceptual understanding.
- Students receive delayed or limited feedback on spoken explanations.
- Frequency: High (Occurs during every assessment cycle.)

9. PROBLEM ROOT / CAUSE [RC]

- Dependence on manual evaluation.
- Lack of intelligent tools for spoken concept assessment.
- Inconsistent evaluation criteria among different faculty members.
- Delayed feedback affecting student improvement.

7. BEHAVIOR + ITS INTENSITY [BE]

- Students regularly participate in viva examinations and presentations.
- Faculty manually evaluate spoken responses.
- Institutions seek scalable assessment methods.
- Intensity: High during examinations and academic evaluations.

3. TRIGGERS TO ACT [TR]

- Need for faster and fair evaluation.
- Increasing use of AI in education.
- Demand for objective concept-based assessments.
- Need to improve students' communication and presentation skills.

4. EMOTIONS BEFORE / AFTER [EM]

Before

- Anxious
- Uncertain
- Stressed
- Lack of confidence

After

- Confident
- Motivated
- Satisfied
- Aware of areas for improvement

10. YOUR SOLUTION [SL]

Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered web application that records or accepts students' spoken responses, converts speech into text, evaluates conceptual understanding using semantic similarity techniques, generates objective scores, provides personalized feedback, and produces downloadable PDF reports. The solution improves assessment accuracy, reduces faculty workload, ensures consistent evaluation, and helps students enhance both conceptual understanding and communication skills.

8. CHANNELS OF BEHAVIOR [CH]

ONLINE

- Streamlit Web Application
- Learning Management Systems (LMS)
- Educational Portals

OFFLINE

- Classroom Viva
 - Laboratory Assessments
- Seminar Presentations